

Context-Aware Linguistic Steganography: A Generative Workflow using Large Language Models

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This paper proposes a black-box linguistic steganography workflow for threaded discussions that hides bits by choosing a reply target and a comment topic, then writing an ordinary comment. The receiver is designed to recover those choices from a supplied thread snapshot and shared operational parameters, without relying on any hidden markup. The method does not use invisible characters. To assess generated comments, a separate language model evaluated 244 paired outputs on several criteria. In the three-reply task, it chose our comment as the least human-like in 29.1% of cases, compared with 47.5% for ZLG, and rated our comments higher on register fit (3.193 versus 2.791). In the ten-comment insertion task, it identified our comment more often than ZLG's (51.6% versus 28.3%). These task-specific judgments do not establish steganographic security or independent payload recovery.

Additional Key Words and Phrases: Linguistic Steganography, Large Language Models, LLMs, Natural Language Processing, NLP, Black-box Steganography, Semantic Embedding, Agentic Workflows, Imperceptibility

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1 INTRODUCTION

Linguistic steganography hides a secret inside ordinary text. Early methods mostly swapped synonyms [1]. Natural language has little spare room, so even small statistical quirks can give a detector something to grab [2–4]. Large language models changed the cover: instead of editing an existing sentence, many systems now generate the text [5–7]. Those systems usually hide bits by steering next-token sampling, which can look fluent while still sitting on a modified distribution.

That fluency still has to fit the thread. A model can write a grammatical, low-perplexity sentence that does not belong in the conversation [8–10]. A comment that is fluent but off-topic is conspicuous to a human reader and to a semantic checker.

We hide the payload in two conversational choices rather than in token probabilities. Bits select (i) which comment to reply to in a depth-first flattened thread, and (ii) which topic the comment will discuss from a list both parties rebuild from the public thread and a shared research procedure. A language model then writes an ordinary visible reply for that pair. We do not use invisible characters. Before keeping the comment, the sender uses the same topic-identification step as the receiver and retries if it identifies the wrong topic. Figure 1 contrasts token-level embedding with this two-layer selection channel.

The primary contributions of this paper are:

- A two-layer selection-channel architecture for threaded discussions that encodes bits through reply targeting and comment topic selection, with a visible reply rather than a hidden carrier.

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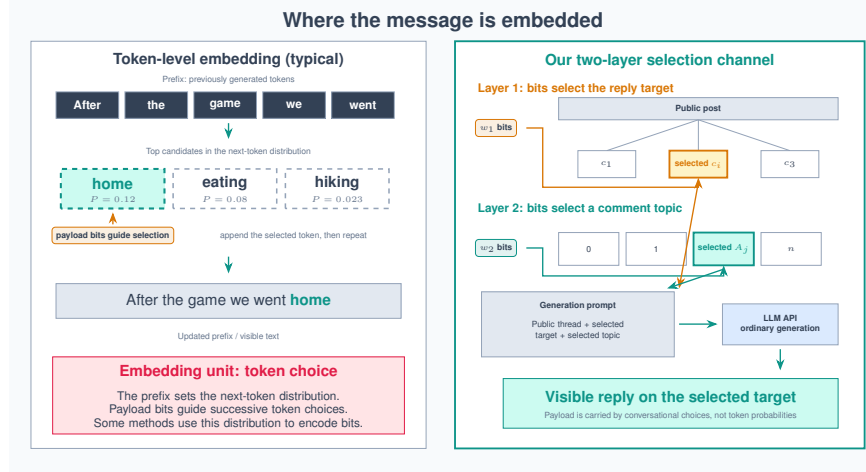


Fig. 1. Conceptual contrast between token-level embedding and our two-layer semantic embedding over a threaded discussion.

- A receiver-side reconstruction pipeline (thread plus hash-checked research) that does not need white-box logits. The reported results do not exercise it alone: they use audit-assisted reconstruction from sender-side metadata.
- A closed-loop verification mechanism that simulates decoding before the comment is kept and retries synthesis, with an optional revision pass, under stochastic generation.
- An independent LLM-judge comparison on the 244-post unique-ZLG subset against the official ZLG baseline, reported with source-run success rates and a historical 304-pair configuration table, without a superiority claim.

2 BACKGROUND

This paper is about hiding that communication is happening, not about encrypting the bytes. **Steganography** embeds a message in an ordinary carrier, such as an image or a piece of text, so an observer has no reason to look for a hidden channel. **Imperceptibility** is the primary objective, not confidentiality as such.

Text is a particularly challenging carrier due to its low redundancy and strict semantic constraints. The classical “Prisoners’ Problem” [11] illustrates the goal: two parties, Alice and Bob, must exchange hidden information without alerting a watchful adversary.

Textual steganography methods are typically divided into **format-based** approaches, which exploit layout or structural features, and **linguistic-based** approaches, which modify linguistic form. Within the latter, early techniques such as **synonym substitution** embed bits by altering lexical choices, but suffer from low capacity and high detectability.

Most published generative methods hide bits by biasing next-token sampling. Given a language model and a secret bitstream, the encoder restricts which token may be emitted at each step so that the choice encodes message bits. That process is often called steganographic sampling [12]. It is the literature’s token-level model, not the proposed method in this paper. Security arguments for those systems turn on keeping the modified sampling distribution close to the model’s original next-token distribution.

Traditional linguistic approaches offer limited embedding capacity and often leave statistical artifacts. Advances in deep learning and **Large Language Models (LLMs)** now enable generative methods that achieve higher text quality

and more secure embedding. Evaluating such systems requires several dimensions of imperceptibility: **perceptual** (human naturalness), **statistical** (distributional similarity to natural text), and **cognitive** (semantic and contextual fidelity) [13].

LLM use in steganography also introduces new sources of detectability. **Hallucination**—where a model generates content that is factually unsupported, semantically inconsistent, or poorly grounded in the preceding context—is one such source. In ordinary text generation, hallucination is mainly a reliability concern. In steganographic generation, it becomes a security concern as well: even a fluent sentence can expose a hidden channel if it contains an unsupported claim, an abrupt topic shift, or a contradiction with the surrounding context. Hallucination therefore threatens **cognitive imperceptibility**, since generated text must remain contextually plausible and communicatively coherent, not merely surface-fluent.

Distinct from hallucination, the **PSIC effect** (Perceptual-Statistical Imperceptibility Conflict) [2] captures a different and arguably more fundamental tension in generative linguistic steganography. Text that is optimized for human fluency is not necessarily statistically indistinguishable from natural human text. For example, selecting only high-probability tokens can produce sentences with low perplexity and strong local readability, but it may also make the generated distribution overly smooth, regular, or concentrated compared with real human writing. Such text can appear natural to readers while remaining detectable to statistical steganalyzers [2].

The PSIC effect therefore shows that **perceptual imperceptibility** and **statistical imperceptibility** are related but not identical goals. Human-written text often contains variation, irregularity, and locally unexpected word choices. By contrast, overly fluent model-generated text may lack this natural variance. Increasing the embedding rate can sometimes force the sampler to select from a broader range of tokens, which may better approximate the diversity of human text; however, this same process can also reduce readability or semantic coherence if the token choices become poorly aligned with the context. Thus, higher capacity does not automatically imply better security. Secure generative steganography requires balancing embedding rate, fluency, distributional similarity, and contextual consistency.

2.1 Large Language Models as Text Generators

Large Language Models (LLMs) predict the next token from a prefix [3, 14, 15]. That is why token-level steganography can ride sampling, and why this paper does not: we never modify the next-token distribution.

2.2 Large Language Models in Generative Linguistic Steganography

Because text produced by modern LLMs can closely resemble ordinary human-authored text, these models provide effective engines for generative text steganography. Rather than editing an existing cover, a growing line of work treats the model itself as a steganographic sampler: the hidden message steers generation, so the stego text is drawn from a realistic communication distribution from the start. The public availability of high-quality models, together with the efficiency of sampling-based embedding, has made this route more practical than earlier designs.

LLMs such as GPT-2 [15], LLaMA [16], and Baichuan2 [17] are common backbones. Most schemes still tie message bits to token choices: at each step the bits pick which candidate is emitted. White-box methods need the exact model and vocabulary, and they change the sampling distribution when they embed [7]. Black-box methods call an API or UI and never see logits. That is the distinction this paper needs. In our method, the sender generates the reply through an ordinary LLM call. To recover Layer 2, the receiver asks an LLM to identify the comment topic without access to the model’s token-probability distribution.

New approaches, such as **LLM-Stega** [7], explore black-box generative text steganography using the user interfaces (UIs) of LLMs, which removes any need to observe internal sampling distributions. Secret bits are encrypted and mapped onto a prearranged keyword set that steers generation. Recovery checks provide feedback: candidates that do not allow message recovery are discarded, preserving both extraction accuracy and semantic quality [7].

Co-Stega retrieves recent posts and uses prompts to raise reply entropy, aiming at higher capacity in social-media settings without dropping fluency [8].

Some earlier white-box methods fine-tune a pretrained model on data from a target dataset and context [2, 3, 9, 18]. By contrast, **zero-shot generative linguistic steganography** does not fine-tune the model on that task-specific data; it relies on in-context learning instead. Covert text samples are supplied in the prompt, and the stego text is produced within a question–answer (QA) exchange, so the output reads as a natural reply rather than as free-standing generated text [6].

The increasing popularity of deep generative models has made it feasible for provably secure steganography to be applied in real-world scenarios, as they fulfill requirements for perfect samplers and explicit data distributions [3, 19, 20].

LLM-based steganographic methods are typically evaluated on two primary axes: imperceptibility (perceptual, statistical, and cognitive measures) and embedding capacity (e.g., bits per token or bits per word). Imperceptibility evaluations may include automatic metrics (PPL, Distinct-n, KL/JSD) as well as human evaluations; embedding capacity is usually reported as bits/token or overall embedding rate. Section 3 surveys how the literature applies, evaluates, and constrains these methods in more detail.

3 RELATED WORK

Majeed et al. (2021) [21] surveyed pre-LLM text steganography techniques, predating the current transformer era, emphasizing synonym replacement, spacing manipulation, and coding schemes such as Huffman coding. The methods of that generation often relied on context-free grammars or Markov chains and tended to produce syntactically correct but semantically incoherent cover text. Setiadi et al. (2025) [22] examine more recent AI-powered data hiding across image, linguistic, and 3D mesh carriers (2021–2025), observing that LLMs have “revitalized” linguistic steganography through methods built on GPT-2 [23], GPT-3 [24], LLaMA2 [25], and Baichuan2 [26]. On the detection side, Wang et al. [4] review deep-learning linguistic steganalysis and document how detection techniques have evolved alongside generative hiding methods.

Those surveys cover the field. None of them treats fit-to-thread as the main criterion for LLM-based methods. We do, because that is the channel. The remainder of this section synthesizes 31 primary studies from 2020 to 2025 on what was published, where it was applied, how it was evaluated, how it used external knowledge, and where it still breaks. Table 1 places this synthesis next to the three surveys above.

3.1 State of the Published Literature

The 31 studies span 2020 through 2025, with 10 publications falling in 2020–2023 and 21 in 2024–2025 alone—a sharp acceleration that coincides with the wider availability of large-scale models. Widely used model families—GPT, GPT-2, and LLaMA [15, 16, 24]—appear across both periods, though proprietary systems grow more common in the later studies as black-box access becomes a viable design choice. Across the reviewed corpus, open-weight and research-accessible models appear more frequently than closed commercial systems, reflecting the practical importance of reproducibility and controllable experimentation.

Table 1. Comparison of the literature synthesis in this section with prior surveys of text/linguistic steganography.

Survey	Period covered	Methodology	Carrier scope	LLM-era coverage	Contextual-compatibility focus
Majeed et al. [21] (2021)	Pre-transformer	Narrative	Text-only (synonym, spacing, coding)	None	No
Setiadi et al. [22] (2025)	2021–2025	Narrative	Multi-carrier (image, linguistic, 3D mesh)	Yes	No
Wang et al. [4] (2023)	Deep-learning era	Narrative	Detection-side (linguistic steganalysis)	Partial	No
This synthesis	2020–2025	Structured synthesis	Text-only (LLM-based)	Yes	Yes

Table 2. Representative model families across the reviewed studies. Categories are non-exclusive, so totals exceed 100%.

Model Type (Studies)	Models and Representative Works
Open-weight Models (25/31, 80.6%)	GPT-2 [3, 9, 18, 19], LLaMA/LLaMA2 [6, 8, 20, 33], BERT [1, 10, 13, 34–39], OPT [40], BART [1, 5], Qwen [30, 41]
Proprietary Models (7/31, 22.6%)	GPT-3.5/4, ChatGPT [7, 27–29, 42, 43]
Custom or Task-Specific Architectures (5/31, 16.1%)	From-scratch or task-specific models [2, 32, 35, 36, 40]

The field has also evolved from earlier white-box token-sampling schemes toward a broader design space that includes hybrid and black-box methods. The reviewed studies now span generative schemes that write stego text from scratch [2, 3, 12, 19], rewriting systems that paraphrase existing covers [5], black-box pipelines that treat the model as an opaque API [7, 27–29], zero-shot protocols driven primarily by prompting [6], context-aware frameworks that retrieve or constrain external information to increase semantic plausibility or payload [8, 9, 30], and methods that emphasize stronger security, robustness, or token-level quality control [3, 19, 31, 32].

Table 2 shows how the reviewed studies distribute across model families. This table is best read as a synthesis of methodological tendencies rather than a strict one-study-one-category classification: many papers combine multiple model families, so the counts overlap and the percentages do not sum to 100%. Open-weight models account for the largest share because they support reproducible experiments and direct control over sampling or decoding, with GPT-2 and LLaMA-family systems appearing especially often. BERT-based methods also recur frequently, but usually as masked-language or scoring components rather than as the sole generation backbone. Proprietary systems such as GPT-3.5/4 are mainly used in black-box or prompt-driven settings, while custom or task-specific architectures appear less often but remain important for studies that prioritize provable security, reversible encoding, or tightly controlled payload design.

3.2 Application Domains

The reviewed studies show that **covert communication** remains the dominant application of LLM-based linguistic steganography. In this setting, secret information is embedded into seemingly benign text so that the message can pass through ordinary communication channels with minimal suspicion. LLM-based methods improve this use case

Table 3. Application categories across the reviewed studies. Categories are non-exclusive.

Primary Application	Variant / Category	Representative Studies	Total Count
Covert Communication	Generation-based (from scratch)	[2, 3, 18–20, 31, 32, 35, 37–39]	11
	Paraphrasing, Rewriting, or Editing	[5, 34]	2
	Black-box and Prompt-based	[6, 7, 27–29, 33]	6
Context-Aware Systems	Domain, Emotion, or Social Adaptation	[8–10, 13, 30, 41]	6
Watermarking	Content Attribution (Boundary Cases)	[1, 36, 40, 42–44]	6

by producing more fluent and contextually plausible outputs than earlier rule-based or local-substitution systems [2, 3, 7].

Within this application space, the reviewed papers explore several operational variants. Some methods generate stego text from scratch using a shared model or sampling strategy [3, 19]. Others rely on paraphrasing or rewriting so that a pre-existing cover text can be transformed while preserving its apparent meaning [5]. Recent black-box and prompt-based approaches such as **LLM-Stega**, **DeepStego**, and multi-criteria linguistic optimization show that covert communication can also be implemented through commercial or hosted interfaces without direct access to token probabilities, using prompt design, keyword sets, stylistic features, and reject-sampling style verification instead [7, 28, 29]. Context-aware systems such as **Co-Stega**, **Hi-Stega**, and emotionally controllable steganography further adapt the generated text to specific domains, topics, entities, emotional tones, or social settings in order to improve plausibility and increase effective payload [8, 9, 30]. This last variant—adapting generation to the surrounding domain and discourse rather than relying on generic fluency—is the closest precedent in the literature to the context-aware, thread-level embedding developed in Section 4.

Although adjacent topics such as watermarking, authorship attribution, or prompt-hiding attacks are relevant to the broader field of information hiding, they remain analytically distinct from covert payload transmission. The 2025 watermarking studies are therefore treated as boundary cases: useful for understanding robustness, semantic preservation, and detection methods, but not interchangeable with steganographic communication systems [44].

A comprehensive breakdown of how the reviewed studies are distributed across these primary application areas and variants is summarized in Table 3.

3.3 Evaluation Practices in the Literature

How published LLM steganography is scored varies by paper, which makes cross-paper comparison hard. Section 6 states the formulas this paper actually computes: recoverable selection-channel width, attempt versus accepted-output rates, GPT-2 perplexity on the historical run, and the five LLM-judge criteria. The rest of this subsection is about what the 31 studies reported, not a second copy of those formulas.

3.3.1 Perplexity. Perplexity is an imperceptibility metric for fluency, where lower values generally indicate more natural text [45]. However, its usefulness for language model evaluation is limited. It mainly reflects internal consistency rather than factuality or reasoning, so fluent but implausible statements can still receive low perplexity scores [46]. It is also sensitive to text length, with short sequences often receiving inflated scores even when they are highly fluent [47]. In addition, perplexity is not directly comparable across models unless tokenization and vocabulary are aligned, since these design choices affect the score itself [48]. Fang et al. [47] further argue that token-averaged likelihood

can under-represent performance on important low-probability tokens, while Wang et al. [46] note that repetition can sometimes lower perplexity in misleading ways. Taken together, these issues suggest that perplexity should be used cautiously and alongside other measures when assessing steganographic text quality.

3.3.2 Statistical Divergence Metrics. Kullback-Leibler Divergence (KLD) [49] and Jensen-Shannon Divergence (JSD) are information-theoretic metrics used to evaluate steganographic security. KLD quantifies information loss by measuring the relative entropy between cover and stego distributions, serving as the theoretical standard for security modeling despite being asymmetric and failing as a strict distance measure. JSD improves upon this as a symmetric, bounded variant that measures how far each distribution lies from their average, providing a more stable basis for formulating statistical imperceptibility bounds—particularly when language models approximate human text distributions. Together, these two metrics attempt to capture how closely steganographic outputs mimic legitimate communication channels.

However, real-world application reveals reliability limits, most notably the PSIC effect [2]. KLD and JSD scores can diverge from human judgment as statistical optimization progresses: methods with favorable divergence scores may still produce low-quality text that appears suspicious to human observers. This discrepancy is also dataset dependent—the same method can yield different divergence values on different corpora at equivalent embedding rates (e.g., KLDs of 19.507 on IMDB versus 8.295 on Twitter for VAE-Stega [2]), making cross-paper comparison unreliable. In addition, studies use different formulas, feature spaces, and measurement scales, including latent BERT features versus direct word distributions. Meteor [3], for example, is associated with reported KLD values ranging from 0.045 in one setting to 7.491–11.845 in others. Consequently, these metrics capture distributional distance but not perceptual naturalness, so they should be paired with semantic and human-centered measures.

3.3.3 Capacity Metrics. Capacity is judged by four metrics:

- **Bits per Token (BPT):**

$$\text{BPT} = \frac{\text{Total Secret Bits}}{\text{Total Tokens}} \quad (1)$$

- **Bits per Word (BPW):**

$$\text{BPW} = \frac{\text{Total Secret Bits}}{\text{Total Words}} \quad (2)$$

- **Embedding Rate (ER) [50]:** Average density of hidden information per textual unit

$$\text{ER} = \frac{1}{N} \sum_{i=1}^N \text{bits}_i \quad (3)$$

where N is the number of textual units (words, tokens, or sentences) and bits_i is the number of bits embedded in the i -th unit. In some fixed-payload schemes, the total number of embedded bits is predetermined for each stego text and does not depend on its length; for example, LLM-Stega can assign a constant payload to every generated stego sentence, so the embedding rate can also be expressed as $\text{ER}_{\text{fixed}} = B_0$, where B_0 is the fixed number of bits embedded in each stego text.

- **Utilization Rate (UR):**

$$H = - \sum_{x \in \mathcal{X}} P(x) \log_2 P(x) \quad (4)$$

$$\text{UR} = \left(\frac{\text{Actual Bits Embedded}}{H} \right) \times 100\% \quad (5)$$

Table 4. Five primary bias categories affecting capacity metrics in steganographic evaluation.

Bias Category	Core Problem	Critical Implication
Tokenization Inconsistencies	Metrics depend entirely on specific tokenizers (e.g., GPT-2 BPE vs. word-level)	Direct comparisons across papers become meaningless when tokenization strategies differ
The PSIC Effect	Conflicts between imperceptibility and statistical security are ignored	High capacity may degrade human fluency while paradoxically improving detection resistance
Model Training Bias	Utilization Rate calculations assume uniform token availability	Actual hiding space is smaller than theoretical entropy due to model frequency preferences
Reporting Ambiguities	No standard definition of “capacity” across systems	Practical payload vs. effective payload distinctions create misleading efficiency claims
Context Blindness	Density metrics treat text as neutral bit containers	Semantic incoherence constitutes a security failure that BPW/BPT fails to penalize

These quantities describe how densely a secret is embedded, but they are affected by systematic biases that limit cross-system comparison, summarized in Table 4.

Tokenization differences make “1 BPT” from one paper difficult to compare with “1 BPT” from another when the studies use different tokenizers. The PSIC effect shows that higher density can hurt human fluency yet help statistical evasion. Model frequency preferences shrink the real alphabet to high-probability tokens, so naive entropy limits overstate usable space. Ambiguous reporting—practical vs. effective payload, or bit length vs. stego length—can make capacity claims appear stronger than they are. Finally, BPW/BPT ignore semantics, so high values can be reported even when the resulting text is suspicious or incoherent. Recent works reveal additional distortions: loop-count overhead, dictionary-size caps, baseline-dependent “improvements,” watermarking goals that invert the desired signal, conversational filler that dilutes BPW, and nonlinear ER curves that make any single threshold difficult to interpret. The added 2025 studies reinforce this point: DeepStego emphasizes high capacity and recovery accuracy, and multi-criteria optimization reports reliable stylistic decoding [28, 29], but these results also make cross-paper comparison harder because the reported gains depend on different baselines, prompts, datasets, and decoding assumptions.

To contextualize theoretical capacity metrics, Table 5 summarizes representative embedding capacities across the reviewed methods. Capacity calculations vary significantly based on the encoding mechanism. **Fixed-length mapping** techniques, such as **LLM-Stega** [7] and **GCStego** [41], derive capacity from rigid keyword or token mappings. LLM-Stega reports 64 bits per sentence. GCStego encodes each word of the secret message as a fixed 16-bit vocabulary index, then hides $\log_2 K$ bits in each substituted cover token, so its 16 bits are a payload-encoding width rather than a cover embedding rate. **ParaLS** [1] operates on a smaller scale; its worked example carries a 3-bit binary watermark (“110”). In contrast, **payload-dependent** methods scale with the cover text or secret message length: **Meteor’s** capacity of 1280 bits corresponds to an embedded payload of 160 bytes, **Zero-shot** capacity is directly tied to the Base64 representation of the secret bitstream, and **Discop** uses a statistical approach that, with a truncation parameter of $p = 0.95$, achieves an entropy rate of 4.84 bits per token, yielding 968 bits over a 200-token sequence.

3.3.4 Security Metrics. The sources evaluate security and reliability through the following integrated metrics:

- **Detection Accuracy (Acc) / Error Rate (PE):** Measures a classifier’s ability to distinguish between cover and stego text. Studies treat an accuracy near 50% (or PE near 50%) as the target, meaning that detector performed at chance under that evaluation. It does not show that stego and normal text are statistically identical; a different detector, feature set, or training split may still separate them.
- **F1 Score:** The harmonic mean of precision and recall, used to report detection reliability. It is most informative when cover and stego classes are imbalanced, where accuracy alone can look high for a detector that favors the majority class.

Table 5. Representative capacity calculations reported by selected methods. Full generated passages are omitted because they do not support direct cross-study comparison.

Method	Reported Capacity	Encoding Basis	Comparability Note
Meteor [3]	1280 bits	160-byte payload $\times$ 8 bits	Payload total; depends on the selected message length.
LLM-Stega [7]	64 bits per sentence	Fixed semantic-slot mapping	Sentence-level capacity; not directly comparable with BPW or BPT.
GCStego [41]	16 bits per secret word	Fixed 16-bit token index per secret-message word	Payload-encoding width, not a cover embedding rate; each substituted cover token carries $\log_2 K$ bits.
ParaLS [1]	3 bits	Binary lexical watermark	Worked-example payload (“110”) rather than a normalized embedding rate.
Zero-shot [6]	Token-level BPW/capacity	Generated-token coding	Capacity depends on the representation, model, and prompt protocol; not directly comparable with selection bits.
Discop [19]	968 bits over 200 tokens	4.84 bits per token at $p = 0.95$	Token-level rate tied to the model, tokenizer, and truncation setting.

- **Robustness (Attack Resistance):** Evaluated via the Mean Impact of Attack (mIOA) or bit accuracy after removal attempts such as denoising auto-encoding, word substitution, or sentence deletion.
- **False Positive Rate (FPR):** Measures how often human or normal model-generated texts are incorrectly flagged as containing secret messages.

Recent robustness-oriented work broadens this evaluation set. Position-agnostic generation tests perturbation tolerance across multiple attack types, active-attack repair methods evaluate message integrity after tampering, and contrastive semantic watermarking reports F1 under rewriting and substitution attacks [31, 32, 44]. This shift suggests that future evaluations should report not only clean-channel capacity and fluency, but also degradation curves under realistic editing.

3.3.5 Evaluation Challenges and Gaps. The broad range of evaluation metrics across the reviewed studies is summarized in Table 6, which categorizes metrics by type and quantifies their usage frequency. The numbers in that table reveal several persistent gaps. Shared benchmarks are rare: only around 20% of studies use common datasets, which makes cross-paper comparison difficult even when the same metric name appears in both papers. Human evaluation was reported in just one of the 31 reviewed studies—a surprisingly low share given how central perceptual plausibility is to the field’s core claims. Robustness testing is similarly sparse: only 5 of the 31 studies report it, so it is absent in roughly 84%. Many papers also evaluate along only one or two metric dimensions, leaving imperceptibility, security, and capacity to be inferred from partial evidence rather than directly measured together. Taken together, these gaps mean the strongest conclusions available from the reviewed corpus are qualitative rather than quantitative. Section 6 reports an independent LLM-judge comparison together with source-run reliability and a historical configuration table; GPT-2 and unigram divergence are scored on that historical run, not on the 2026-08-15 freeze.

3.4 Integration of External Knowledge Sources

External knowledge integration is a recurring design pattern in the reviewed literature. Across the surveyed studies, external information serves three main purposes: improving contextual plausibility, maintaining topic consistency, and—in some cases—expanding the effective space available for embedding. Table 7 summarizes the main knowledge-integration patterns and their roles.

Rather than relying only on the internal prior of the language model, several recent systems inject external guidance during generation. Retrieved posts, headlines, or comments can provide a realistic discourse context for the cover text [9]. Knowledge-graph triples can constrain the content that should appear in the output, improving semantic

Table 6. Frequency of evaluation metric categories in the reviewed studies. Categories are non-exclusive.

Dimension	Metric Category	Studies	Main Comparability Constraint
Imperceptibility	Perplexity	13	Model, tokenizer, corpus, and sequence length
Imperceptibility	Statistical divergence (KLD/JSD)	10	Estimated distributions and sampling settings
Imperceptibility	Semantic similarity	13	Reference construction and metric choice
Imperceptibility	Human evaluation	1	Protocol, annotator population, and sample size
Security	Detection accuracy or F1	21	Detector, class balance, and train/test data
Capacity	BPW or BPT	27	Tokenization and payload accounting
Robustness	Recovery or attack resilience	5	Attack type, strength, and evaluation protocol
Efficiency	Generation or extraction time	5	Hardware, model size, and implementation

Table 7. Representative patterns of external knowledge integration.

Knowledge Type	Typical Role	Primary Benefit	Examples
Semantic Resources	Topic, entity, and emotion guidance	Better semantic control	Co-Stega [8], knowledge-graph guided methods [33, 35], emotionally controllable steganography [30]
Domain Corpora	Domain adaptation	Better stylistic fit	FREmax [18], specialized datasets
Prompt Engineering	In-context control	Better task alignment	Zero-shot [6], DeepStego [28], multi-criteria optimization [29]
Context Retrieval	Evidence or example retrieval	Higher contextual plausibility	Co-Stega [8], retrieval-augmented pipelines [9]

consistency over longer spans [33]. Named entities and sentiment labels add a finer-grained control layer when the goal is emotional or conversational consistency [30]. Frequency-based external statistics can also be used to steer generation toward distributions that better match the target channel [18].

Another recurring pattern is the use of domain-specific corpora. Fine-tuning or conditioning on domain data helps align the generated text with the vocabulary, stylistic conventions, and topical expectations of the intended communication channel. This pattern appears in systems such as VAE-Stega [2], Meteor [3], Hi-Stega [9], FREmax [18], Rewriting-Stego [5], and Summarization-Stego [10]. When additional training is not possible, black-box methods move this domain adaptation into the prompt, style specification, or retrieval stage. Zero-shot approaches provide examples directly in the context window, LLM-Stega frames the generation request with topic-constrained prompts, DeepStego uses prompts and synonym guidance, multi-criteria optimization encodes through stylistic controls, and Co-Stega retrieves recent posts to anchor the generated reply in a plausible local context [6–8, 28, 29]—the same retrieve-then-anchor strategy that underlies the shared research stage described in Section 4.2.

Table 8 provides a more detailed breakdown of how external knowledge sources are integrated across the reviewed studies.

These five patterns differ primarily in how directly they constrain the generation process. Structured knowledge approaches [33, 35] use external knowledge graphs or graph attention networks to enforce long-range semantic consistency—a stronger form of control than prompting alone, but one that requires the knowledge resource to exist and be aligned with the target domain. Information retrieval frameworks [8, 9] pull contextually relevant material from external sources at runtime, grounding the cover text in real discourse without requiring model retraining. Linguistic

Table 8. Comprehensive overview of external knowledge integration patterns in LLM-based steganography.

Approach	Variant / Category	Representative Studies	Count
Structured Knowledge Integration	Knowledge Graphs (KG) & Graph Attention Networks	Semantic controllable steganography via knowledge graph [33], joint linguistic steganography with BERT and graph attention networks [35]	2
	Information Retrieval (IR): Retrieval-Augmented Generation (RAG) & Corpus Search	Hi-Stega [9], Co-Stega [8], emotionally controllable text steganography [30]	3
Linguistic Resource Integration	Paraphrase DBs, Synonym DBs & Dependency Datasets	Paraphraser-based lexical substitution [1], DeepStego [28], post-hoc watermarking of black-box LLM text [43], group-chat imperceptible text steganography [41]	4
	External Statistical Analysis: Character/Token Frequency Distributions	FREmax [18]	1
External Model Feedback	Auxiliary Scorer, Critic & Discriminator Models	Rewriting-Stego [5], principled natural language watermarking [36], CPG-LS [37], controllable semantic steganography via summarization [10], ALiSa [39], DeepTextMark [40], contrastive semantic watermarking [44]	7
Shared Knowledge / Side Information	Shared Auxiliary Datasets & Public Contexts	Meteor [3], zero-shot generative linguistic steganography [6], reversible data hiding via masked language modeling [34], natural language steganography by ChatGPT [27]	4

resource methods [1] rely on synonym databases and paraphrase corpora to constrain lexical choice while preserving meaning—a lighter form of intervention that nonetheless depends on the quality and coverage of the external resource. External model feedback is the most common pattern, appearing in seven of the 31 reviewed studies: auxiliary models such as BERT serve as discriminators or soft scorers to assess naturalness or security during generation [37, 40]. Finally, shared side information methods [3, 6] use public models or auxiliary history to synchronize probability distributions between sender and receiver without explicitly passing knowledge through the channel, trading interpretability for coordination efficiency—the same trade-off that motivates the shared search pipeline in Section 4.2.

3.5 Limitations and Trade-offs

Across the surveyed studies, the main constraints recur around imperceptibility, effective payload, semantic coherence, tokenization reliability, deployment assumptions, and attack robustness.

3.5.1 The PSIC Effect Revisited. Generative steganography faces an intrinsic tension: optimizing for perceptual fidelity degrades statistical security. Aggressive perplexity minimization produces unnaturally “sharp” distributions that deviate from high-variance human writing, increasing detection vulnerability. Higher embedding rates force selection

of lower-probability tokens, degrading text quality but expanding the candidate pool to enhance anti-steganalysis resistance. At elevated capacities, output distributions asymptotically approach natural language’s high-entropy chaos, camouflaging signals within legitimate statistical noise [2].

3.5.2 Attack Vulnerability and Security Concerns. Current techniques exhibit a substantial attack surface across multiple vectors, including word-level deletions and lexical substitutions that disrupt dependency parsing. Polishing and re-translation attacks progressively erase watermarks; at high intensities, these structural changes can reduce detection performance to an F1 score of 0.5, rendering the watermark undetectable. Furthermore, adversarial machine learning enables the extraction of secret model parameters from black-box systems, potentially compromising integrity entirely. Recent work responds by moving robustness into the embedding design itself: position-agnostic generation avoids dependence on fixed token locations, active-attack repair models attempt to recover damaged stegotext, and contrastive semantic watermarking tests resilience under rewriting and substitution [31, 32, 44]. To mitigate distributional risks, Variational Auto-Encoders are also employed to learn the statistical distribution of normal text, aiming to reduce the KL divergence between cover and stego distributions and resolve the conflict between perceptual and statistical imperceptibility [43].

3.5.3 Capacity and Contextual Constraints. Short, low-entropy contexts severely constrain embedding capacity. Social media posts and technical documents offer minimal redundancy for hiding [8], while semantic preservation requirements compete directly with information embedding objectives. Brief texts further lack sufficient contextual support for effective steganographic operations [13]. Newer approaches attempt to recover some capacity through synonym guidance and stylistic feature control, but these gains remain tied to the selected prompt, cover text, or candidate-token space [28, 29].

Early generative methods also produced fluent but semantically arbitrary text, violating cognitive imperceptibility [13]. When generation is conditioned only on local token history, the output may drift away from the source topic or conversational logic, which is especially suspicious in low-entropy channels such as social media replies. This contextual drift worsens when methods must keep generating until an arbitrarily large payload is fully embedded: texts become unnaturally long, coherence decays over time, and practical payload efficiency drops even when reported bit capacity appears high [27, 33, 37, 42].

Our paired comparisons against the zero-shot generative linguistic steganography baseline [6] are reported in Section 6. An independent LLM-judge evaluation on the 244-post unique-ZLG subset is the current quality table; a 2026-07-30 configuration comparison reports recoverable capacity and GPT-2 perplexity on 100 post clusters. Both are conditional on accepted outputs, use audit-assisted reconstruction from sender-side metadata on the project side, and do not establish a security advantage for either method.

3.5.4 Segmentation and Tokenization Barriers. Subword tokenization introduces critical extraction ambiguities that are fundamental to white-box steganographic methods. Byte-pair encoding splits words unpredictably, creating multiple valid segmentations that corrupt message boundaries and cause decoding failures. These ambiguities impose hard capacity caps regardless of algorithmic sophistication—SparSamp suffers accuracy degradation from token ambiguity, while ShiMer cannot effectively boost entropy due to tokenization constraints [20].

This segmentation ambiguity persists even when sender and receiver employ identical LLMs and tokenizers. Because modern LLMs decompose words into subword units, and vocabularies frequently contain both complete words

and their constituent components, any given text string admits multiple valid token representations. This multiplicity is compounded by the detokenization-retokenization gap: senders must convert tokens to plain text to conceal their transmission, while receivers must subsequently retokenize this text to recover hidden messages. Despite shared tokenization rules, receivers may select different token sequences than senders intended—a sender might encode a message using two separate tokens (“any” and “thing”), only for the receiver to retokenize the surface string as the single token “anything”. Because LLMs operate autoregressively, a single divergent token choice alters the probability distribution for all subsequent predictions, causing total decoding failure for the remainder of the message. To address this white-box-specific challenge, researchers propose SyncPool, a mechanism that aggregates ambiguous tokens into pools and employs synchronized random number generators to ensure both parties select identical tokens [20]. Because this work embeds payload through reply targeting and comment topic selection rather than through raw token identity (Section 4.4.1, Section 4.4.2), it does not inherit that detokenization gap. Layer 1 is a discrete parent index. Layer 2 is recovered by a semantic shortlist and an identification prompt, not by retokenizing a steganographic sampler’s output.

3.5.5 White-box versus Black-box Trade-offs. The architectural choice between white-box and black-box access paradigms involves fundamental trade-offs. Across the reviewed corpus, only a minority of studies operate primarily through black-box or prompt-driven access (Table 3), while the majority employ white-box or hybrid methods. White-box methods typically assume access to internal language-model components such as token probabilities, decoding states, vocabularies, or model parameters, and some approaches additionally require expensive fine-tuning or retraining procedures to align the model distribution with steganographic objectives. While these methods can achieve strong statistical imperceptibility through distribution-aware generation (e.g., VAE-Stega and arithmetic-coding-based methods), they suffer from reduced accessibility, high computational cost, and deployment complexity.

In contrast, black-box approaches operate through standard LLM APIs or user interfaces without requiring direct access to model internals or retraining, substantially improving practicality and usability. However, the distinction is primarily about *model access*, not the absence of probabilistic coordination, as several black-box approaches still rely on shared probabilistic mechanisms, semantic distributions, or synchronized auxiliary structures between sender and receiver. Recent semantic black-box methods also prioritize robustness against paraphrasing, token perturbation, and noisy communication channels rather than purely token-level imperceptibility. Among the few peer-reviewed black-box methods currently available, LLM-Stega reports a capacity of 5.93 bpw with perplexity 165.76, illustrating that improved accessibility does not eliminate the PSIC trade-off between payload capacity, text quality, robustness, and security. Several recent systems further blur strict categorical boundaries by combining retrieval-, generation-, editing-, or substitution-based paradigms within hybrid steganographic frameworks.

3.5.6 Computational and Resource Constraints. Performance optimization conflicts directly with computational efficiency [3]. Superior results demand increased resources and memory, particularly for large models with external knowledge bases [19]. Real-time latency constraints limit optimization depth, while performance degradation emerges at operational scale. Variable length sampling, rejection sampling, active repair, and token-level quality control significantly increase encoding or decoding complexity, and LLM-based steganography requires 3x to 5x more time than prior methods [20]. The Meteor system exemplifies hardware constraints, running nearly 50× slower on mobile devices compared to GPU environments. LLM-specific deployment obstacles compound these costs: hallucinations and non-sensical continuations can corrupt the covert bitstream or create detectable artifacts, while lossy transmission, editing, or platform-side normalization can irreversibly damage extraction. As model quality improves, steganalysis systems also improve, producing an arms race in which better generation does not automatically translate into better security.

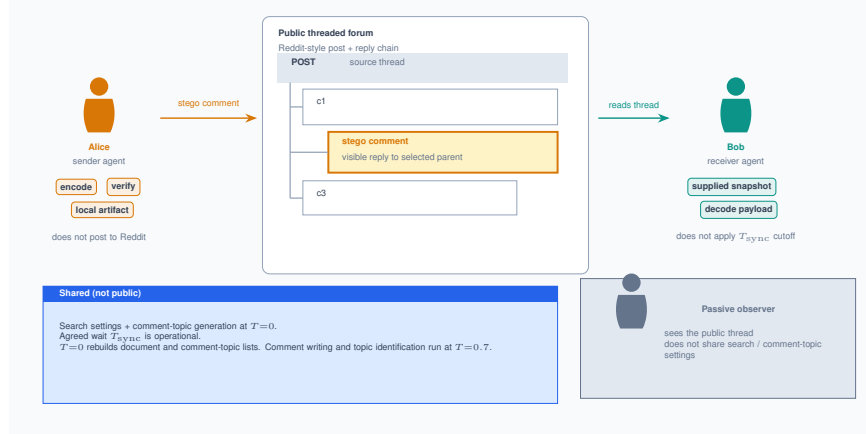


Fig. 2. Covert communication scenario: Alice and Bob use a public threaded channel. They share search and comment topic generation settings, and they agree on a temporal window. The live receiver still needs that snapshot to be supplied; it does not apply the cutoff itself.

3.5.7 Critical Unresolved Challenges. Foundational gaps persist across the field: theoretical security guarantees remain unestablished, resilience to advanced attacks is limited, evaluation frameworks lack standardization, responsible development guidelines are absent, non-English performance lags substantially, and real-world deployment testing remains insufficient. Several studies also surface unresolved concerns around reversibility, bias, and misuse potential, indicating that robust engineering alone will not be enough without clearer security and governance models [1, 6, 34, 51].

Quantitative evidence underscores several of these constraints. The PSIC effect creates a trade-off where increasing the embedding rate (e.g., to ~ 5 bpw) strengthens anti-steganalysis but significantly degrades perceptual text quality [2]. Attack vulnerability is notable; for example, adaptive attacks can reduce bit accuracy to near-random levels (~ 0.6) [36]. Segmentation ambiguity and entropy variability in communication channels are major impediments that lead to decoding failures and restricted practical capacity [3]. Finally, computational optimizations induce significant overhead, with some reordering algorithms causing up to 69% performance degradation in processing speed [19].

3.6 Contextual Compatibility as a Research Direction

The nearest precedents are context-aware systems such as Co-Stega and Hi-Stega, which retrieve or constrain external information so the cover fits a domain. This paper uses that context as the payload channel, not as extra fluency: bits choose a reply target and a comment topic, and the generated comment is an ordinary reply in the thread (Section 4).

4 METHODOLOGY

We propose a black-box steganographic method that hides a secret message inside a public conversation. The sender and the receiver both have accounts on a social media platform. The sender finds a thread, writes what looks like a normal comment or reply, and publishes it. The receiver later reads that same thread, finds the sender’s contribution, and recovers the hidden payload. Our study uses Reddit-style data as the main case. The design targets any threaded discussion environment; whether it transfers cleanly to other platforms remains to be evaluated. Figure 2 shows the end-to-end covert scenario and the shared assumptions needed for synchronization.

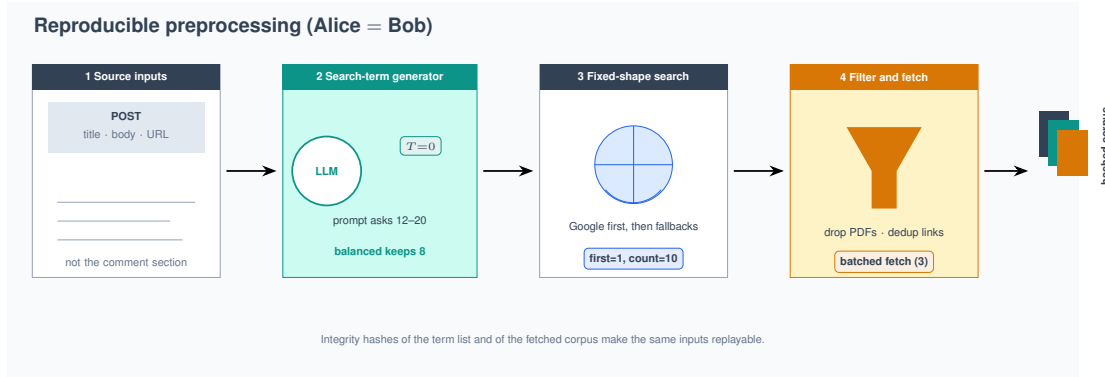


Fig. 3. Research stage: temperature-0 term generation from title, body, and URL; a fixed search request shape; filtering and batched fetching; integrity hashes that detect drift between runs but do not prevent it.

4.1 What Alice and Bob share

Both parties need accounts on the same threaded platform. They also share the search settings, the model used to generate search terms and comment topics at temperature=0, and an agreed wait T_{sync} between Alice's snapshot and the moment she publishes. Temperature 0 applies to reconstructing the document list and the candidate comment topic list. It does not apply to writing the comment or identifying its topic, both of which run at the shared stego temperature (0.7 in the balanced profile used for the reported runs).

4.2 Building a shared document list

Alice's agent picks a busy thread. It then builds a document list that Bob can rebuild: the post, the comments Alice saw, and pages retrieved by a shared search procedure. Figure 3 summarizes that research stage.

Search terms come from the post title, body, and URL, not from the comment section. A temperature-0 prompt asks for at least 12 unique queries and aims for 15–20. The balanced profile then keeps 8 of them. Each kept query is sent with a fixed request shape (first=1, count=10). The live backend tries Google Custom Search first, then other engines if that path is empty. Duplicate links and PDFs are dropped, surviving URLs are fetched in batches of three, and the pipeline records hashes of the term list and of the fetched corpus. Temperature 0 stabilizes the model stages, not the web: search rankings and page contents can change between Alice's retrieval and Bob's. The hashes let Bob detect that drift. They do not make his independent retrieval return Alice's material, and a mismatch there changes the comment topic list that Layer 2 depends on. The reported workloads sidestep this by decoding against the sender's stored topic list (Section 4.3).

From that corpus, the topic extraction step, still at temperature 0, builds an ordered list of candidate comment topics. Each candidate contains a category, a source quote, and the subject the reply should discuss. The reported runs cap the list at 32. Comment-topic extraction is a prior stage. The single-comment encoder does not rebuild the list after choosing a reply target. The reported artifacts used a context-weighted sampler once per post at preparation time, usually with no selected parent. They did not rebuild the topic list after Layer 1 at encode time.

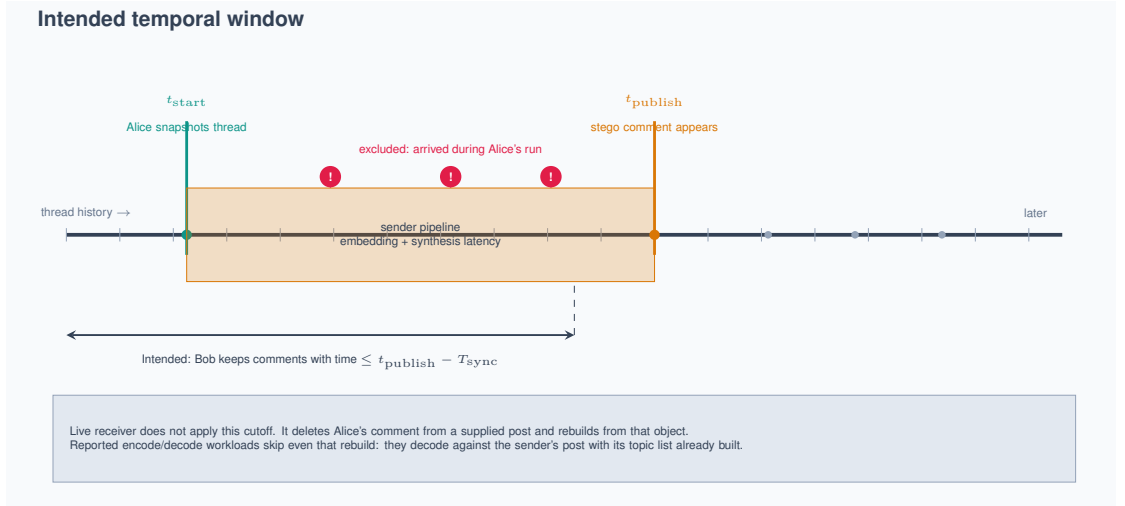


Fig. 4. Intended temporal window: Bob would keep comments dated at most $t_{publish} - T_{sync}$. The live receiver does not apply this filter; it rebuilds from the supplied post after deleting Alice's comment.

4.3 Waiting to publish

Alice needs time to research, pick a reply target, choose a comment topic, and write a comment. If new comments arrive in that interval, Bob cannot be sure which parents Alice saw. The intended fix is an agreed duration T_{sync} : Alice snapshots the thread at t_{start} and publishes at $t_{publish}$; Bob keeps comments with time at most $t_{publish} - T_{sync}$. Figure 4 shows that intended window.

The live receiver does not apply this cutoff. It locates Alice's comment, deletes that subtree from a supplied post object, and rebuilds from what it was given. Supplying a snapshot that already matches Alice's view is an operational duty of the caller. The reported encode/decode workloads skip even that rebuild: they decode against the sender's post with its topic list already built.

4.4 Two choices carry the payload

The payload is not hidden in invisible characters, and it is not hidden by biasing next-token sampling. Bits choose two ordinary visible things: which comment Alice replies to, and what topic her comment discusses. A language model then writes a short public reply for that pair. Figure 5 shows the split.

4.4.1 Which comment to reply to. The nested comment tree is flattened with a deterministic depth-first walk, so both parties number the same comments the same way. There are $C = n + 1$ reply targets: index 0 is a top-level reply to the post, and indices 1 through n are the flattened comments. If the index is not 0, the encoder also builds the parent chain of that comment so the writing prompt can see the local thread.

4.4.2 Which topic the comment discusses. The candidate comment topic list from the research stage is flattened and given sequential idx values. The next bits pick one index. At writing time, the selected topic's category, source quote, and description are added to the prompt with the post, parent chain, and research excerpt matched to the source quote.

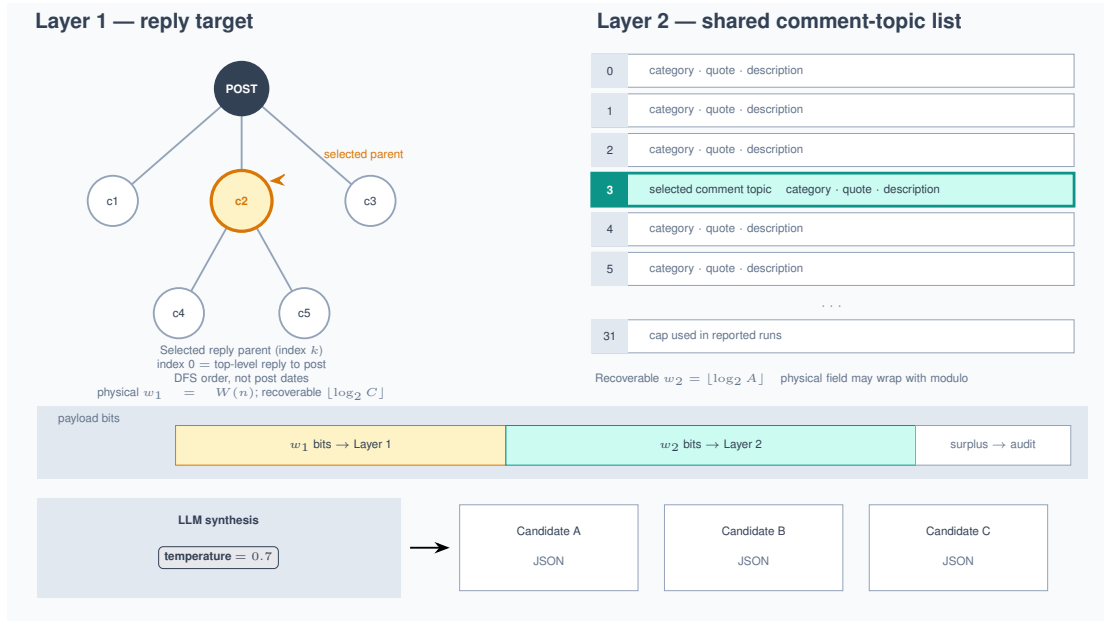


Fig. 5. Two-layer embedding: Layer 1 selects a reply target in a depth-first flattened thread; Layer 2 selects a comment topic from a shared list; an LLM then writes a visible reply for that pair.

4.4.3 How many bits those choices can carry. Both layers consume a prefix of the same bitstream produced in Section 4.7. Layer 1 reads the first w_1 bits; Layer 2 reads the next w_2 .

Two widths must not be mixed. The *physical* widths are the field sizes the single-comment encoder actually writes: $w_1 = W(n)$ and $w_2 = W(A - 1)$, where $W(m) = \lceil \log_2(m + 1) \rceil$ for $m > 0$. If the integer from those bits is larger than the last valid choice, the encoder reduces it modulo the choice count. Generation still succeeds, but several bit patterns then land on the same parent or the same topic, so those patterns cannot be inverted.

The safe fixed-width budget for two independently coded power-of-two subsets is $\lfloor \log_2 C \rfloor$ and $\lfloor \log_2 A \rfloor$. For example, $C = 3$ and $A = 5$ give $1 + 2 = 3$ distinguishable bits; the remaining physical index states alias under modulo. This budget is neither Shannon capacity nor achieved useful throughput. The lossless writer described below makes these selections injective, but the reported single-comment runs used the modulo writer and sender-audit reconstruction. Section 6 distinguishes this available fixed-width selection budget from payload presented and payload independently verified by the receiver.

The live single-comment encoder writes the physical widths and wraps with modulo. A separate lossless writer restricts each field to the recoverable width, then stores that value at the physical width. That injective path is used by the multi-frame planner in Section 4.8, not by the single-comment path that produced the reported tables.

Leftover bits beyond $w_1 + w_2$ are reported as unembedded. They are not dropped, and they are not in the visible comment. Reconstructing the full compressed string from one comment then needs sender-side metadata. We call this audit-assisted reconstruction, and it is what the reported workloads use; we reserve “recovery” for bits Bob decodes from the channel itself.

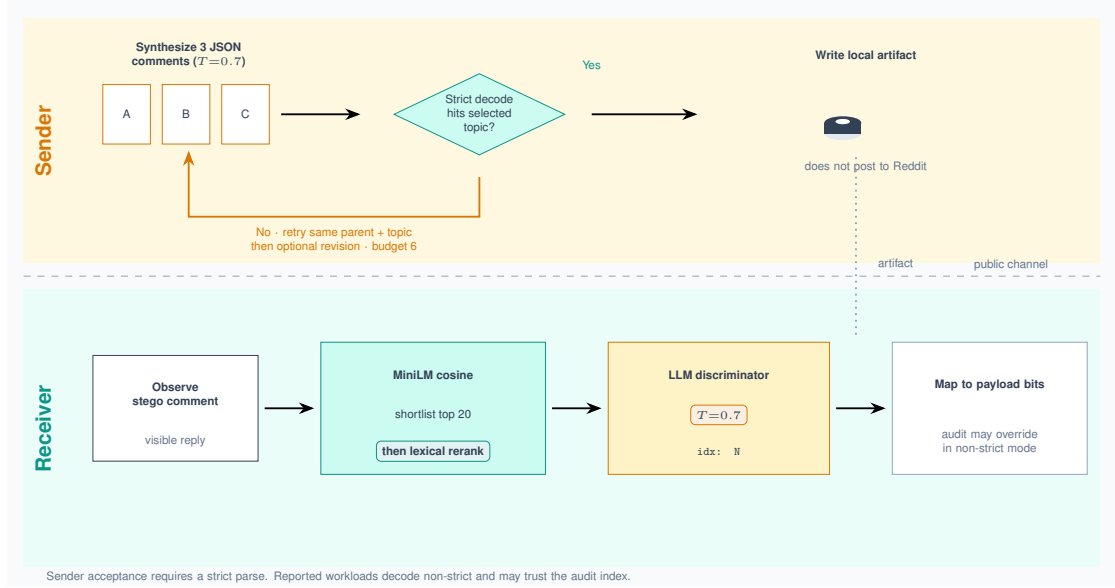


Fig. 6. Sender-side verification retries synthesis until a strict decode identifies the selected topic. The receiver recovers Layer 2 with a cosine shortlist of 20, a lexical rerank, and a temperature-0.7 idx: N prompt.

4.5 Writing the comment and checking that Bob can read it

A nonzero-temperature model writes a JSON array of exactly three short reply candidates. The balanced profile uses temperature 0.7. The prompt sees the post, the parent chain, the selected topic, and a research excerpt. The topic guides the reply; the comment should not announce it as a label.

Before anything is treated as published, the sender runs the same topic-identification step Bob will use, in strict mode. A candidate is accepted only if that step returns the selected topic index and a contextuality gate passes. If none of the three candidates is accepted, the encoder retries with the same parent and topic; the balanced budget is 6 retries after the first attempt. On the last attempt it may also revise a near-miss with a contextual revision prompt rather than sampling a fresh triple. The live sender writes a local artifact. It does not post to Reddit. Figure 6 shows the verify loop next to Bob's recovery path.

4.6 How Bob recovers the bits

Layer 1 is public: Bob sees which comment Alice replied to, and that parent maps back to the DFS index.

Layer 2 is not printed as a number. The full reconstruction path locates Alice's comment by author, deletes it from the supplied post, optionally rebuilds research and the candidate topics, then:

- (1) embeds the comment with a sentence transformer (all-MiniLM-L6-v2) and shortlists the nearest candidate topics, default 20 after a pool of up to 120;
- (2) reranks that shortlist with a mix of cosine similarity and lexical overlap;
- (3) asks the same stego model, at temperature 0.7, to return a single line idx: N naming the canonical topic index.

If the model returns a 1-based rank into the shortlist, that rank is mapped back. If nothing parses, the top semantic candidate is used. Sender acceptance requires a strict parse and rejects those fallbacks. The reported workloads call the

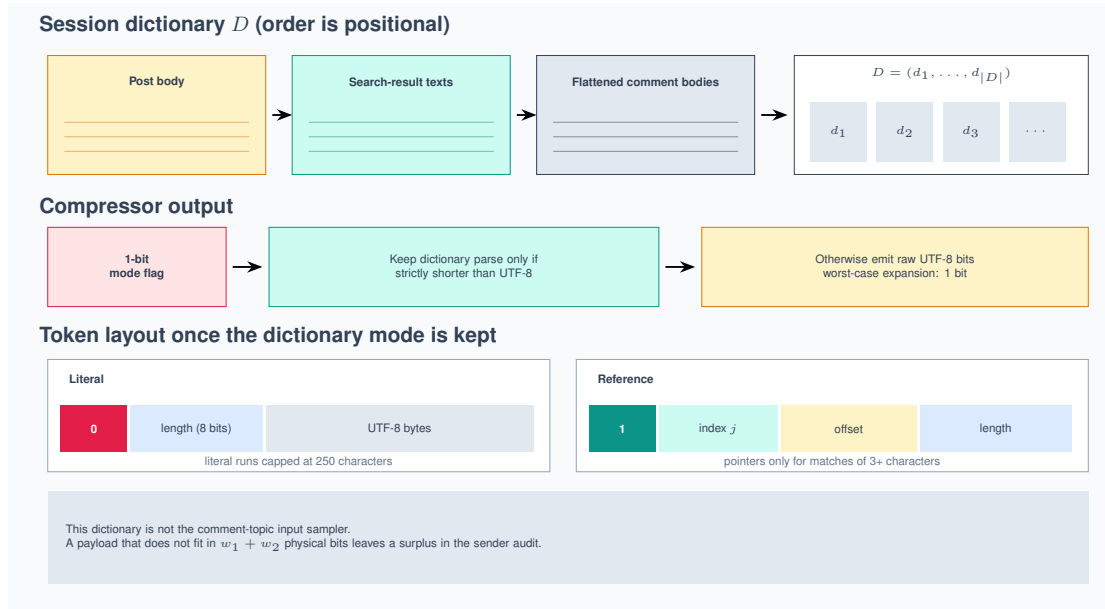


Fig. 7. Payload compression builds a session dictionary from thread and research text, then keeps a dictionary parse only when it is strictly shorter than UTF-8.

decoder in non-strict mode, pass the compressed bitstring from the encode audit, and may treat the audit's topic index as authoritative when it disagrees with the discriminator. Their success counts therefore measure audit-assisted reconstruction, not the receiver described in this subsection. Sender-side acceptance still required one strict discriminator hit at encode time, but no reported number measures Bob decoding Layer 2 from the public thread alone.

4.7 Preparing a text payload

If the secret is text, it is first optionally wrapped by a protection transform, then compressed against a session dictionary built from the post body, the search-result texts, and the flattened comment bodies, in that order. The default transform on the balanced profile is plain: the payload is unchanged. Two other modes exist, `hmac_xor_v1` and `secure_compact_v2`, which add a keyed stream cipher and a truncated MAC; they are recorded on the sender audit so Bob applies the matching inverse. They add confidentiality on top of the covert channel. They are not what makes the channel covert.

The compressor emits a one-bit mode flag, then either raw UTF-8 bits or a dictionary parse of literals and pointers. It keeps the dictionary parse only when that parse is strictly shorter, so the worst-case expansion is one bit. Literal runs are capped at 250 characters; pointers are used only for matches of three characters or more. Field widths and the backward parse that chooses the token sequence are defined once in the shared codec; Section 5 states the token layout. This compressor dictionary is not the same object as the comment topic sampler. Figure 7 sketches the two encoding modes.

The first bit Layer 1 reads is therefore that mode flag. A payload that does not fit in $w_1 + w_2$ physical bits leaves a surplus in the audit, as in Section 4.4.3.

4.8 When one comment is not enough

A longer payload can be split across several frames: each frame is one reply under one post. That planner is implemented. It is not the pipeline behind the reported comparison and ZLG tables, which used the single-comment encoder plus audit-assisted reconstruction.

On the multi-frame path the payload is protected, then compressed against an empty dictionary, which forces the UTF-8 mode of Section 4.7. An Elias-gamma header then stores the frame count F and the payload bit length. Each slot is sized by recoverable capacity from Section 4.4.3, the last slot is zero-padded, and the receiver rejects a stream whose declared F or padding is wrong. This path uses the lossless writer, so the bits-to-selection map is injective. If the sampler is context-weighted, this is also the path that rebuilds candidate topics after resolving the parent of each frame. The receiver does not discover frames in the public thread; the caller must supply an ordered list of comment references.

5 IMPLEMENTATION

The runnable pipeline is five stages: data load, research, comment topic generation, stego encoding, and decode. Each stage writes artifacts to disk so later stages can reuse the same corpus and hashes. Search-term generation and comment topic extraction run at temperature=0 from the same inputs. Comment writing and comment topic identification do not: both use the shared stego temperature (0.7 on the balanced profile).

5.1 Pipeline stages

- Data load fetches unresolved URL content and replaces the post body with the extracted text.
- Research generates search terms from the post title, body, and URL at temperature 0, trims them to the profile cap (8 on balanced), queries a fixed-shape search backend with fallbacks after Google, drops duplicate links and PDFs, and fetches page text in batches of three. It stores the flat search-results corpus and integrity hashes.
- Comment-topic generation flattens the chosen input documents and extracts triplets of category, source quote, and description. Long inputs are split into 30k-character segments and batched up to 80k characters by default. If the backend path fails, the pipeline falls back to a direct LLM prompt with the same JSON schema. The reported runs used a context-weighted sampler at this stage, once per post.
- Encoding protects and compresses the payload, spends bits on a reply target and a comment topic, writes three candidate comments, and keeps a candidate only if a strict local decode identifies the selected topic and a contextuality gate passes.
- Decode shortlists comment topics with MiniLM cosine similarity, reranks with lexical overlap, and asks for idx: N at the stego temperature. The full reconstruction path can rebuild research and comment topics; the reported workloads decode against the sender's post with its topic list already built and pass the audit bitstring.

Figure 8 aligns these stages with the on-disk artifacts.

5.2 Shared codec

The bit layer in Sections 4.7–4.8 lives in one module of pure functions. Dictionary construction for compression, the field-width function W , the token grammar, and the physical-versus-recoverable width rules each have one definition there. The sender and receiver call that module rather than reimplementing it, so round-trip tests can drive the codec with fixtures and no model.

Table 9. Token layout of the compressed payload stream. Every field has a fixed width given the session dictionary D , but decoding is sequential: a token’s start position depends on where the previous token ended.

Token	Field layout	Cost (bits)
Literal	0 length ℓ (8 bits) UTF-8 bytes	$9 + 8b$
Reference	1 index j ($W(D)$ b) offset ($W(d_j)$ b) length ($W(L_{\max})$ b)	$1 + W(D) + W(d_j) + W(L_{\max})$

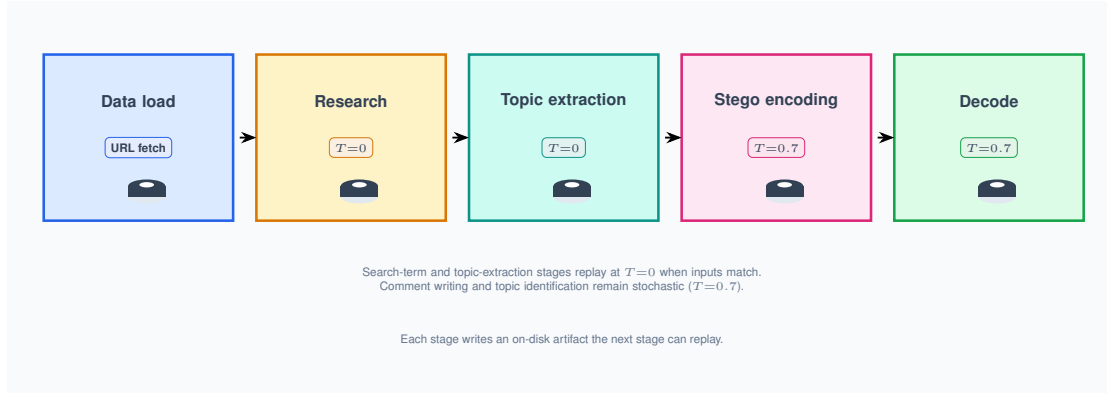


Fig. 8. Implementation stages and persisted artifacts. Search-term and comment topic stages replay at temperature 0 when the inputs match; comment writing and comment topic identification remain stochastic.

The pipelines around it still own parent lookup, comment topic regeneration, semantic search, the discriminator, audit override, and revision. They are not limited to I/O.

Three configuration choices change the bit layer, so both sides must agree: the payload transform (plain, hmac_xor_v1, or secure_compact_v2), the pre-shared secret required by the keyed modes, and whether the compressor’s dictionary capacity profile is on. The sender records the transform in the run audit. The compressor capacity profile is off by default because dictionary indices are positional.

Table 9 is the token layout used by the dictionary parse. A literal is flag 0, an 8-bit character length, then UTF-8 bytes. A reference is flag 1 plus index, offset, and length fields sized by W . Decoding is sequential: a token’s start depends on where the previous token ended.

The single-comment encoder writes physical widths and wraps overflow with modulo. The multi-frame planner uses the lossless writer that stores only recoverable widths.

5.3 Model roles

Exact model IDs are configuration. The roles are not:

- Search-term generation and comment topic extraction run at temperature=0 so both parties can rebuild the same lists from the same inputs.
- Comment writing runs at nonzero temperature (0.7 on balanced) so retries produce different wording for the same parent and comment topic.
- Comment-topic identification uses MiniLM cosine similarity, a default shortlist of 20 with lexical rerank, then the same stego temperature for an idx: N prompt. It is not a temperature-0 oracle.

6 EVALUATION

We compared our comments with the official zero-shot generative linguistic steganography (ZLG) baseline on two separate runs. Table 10 is a quality check. A language model scored a 244-post subset of a 2026-08-15 generation freeze, chosen where ZLG’s output was unique. Table 11 is a 2026-07-30 configuration comparison on capacity, GPT-2 perplexity, length, lexical quality, and BERTScore. The two tables do not share a sample and do not measure the same thing.

Two setup facts apply to every number below. First, none of them comes from the public-thread receiver described in Section 4. Project-side payloads were rebuilt from the sender’s audit metadata (audit-assisted reconstruction), and the decoder was allowed to take the audit’s comment topic index over the discriminator’s. The ZLG reveal lane was token-assisted: it received the sender’s token IDs and an externally supplied payload length, so it does not represent visible-text-only recovery. Second, the run behind the 2026-08-15 comparison set `max_retries` to 1, not the balanced-profile budget of 6 retries after the first attempt that Section 4 describes, so its success rate describes a lower-retry configuration.

6.1 How the printed metrics are computed

The available fixed-width selection budget for two independently coded power-of-two subsets is $3 + 5 = 8$ bits when there are 8 reply targets and 32 candidate topics. In general this safe budget is

$$\lfloor \log_2 C \rfloor + \lfloor \log_2 A \rfloor,$$

where C is the number of reply targets (flattened comments plus the post) and A is the number of candidate comment topics. For non-power-of-two choice counts, leftover physical index states are excluded; for example, $C = 3, A = 5$ yield $1 + 2 = 3$ distinguishable bits, while modulo aliases the remaining states. This is a fixed-width selection budget, not Shannon capacity or achieved useful throughput. On the 2026-08-15 freeze, the stored 13.42 bits/comment for our method is this available budget, not a measurement of independently embedded or recovered useful payload. The audit-assisted reconstruction does not establish receiver recovery. The ZLG values describe its reported successful payload in a capacity-matched setting, not its native token-channel maximum, and its reveal used sender token IDs plus externally supplied payload length.

Success rates count every attempted sample, including failures. Quality scores, including the judge table, are computed only on accepted generations. Failed attempts stay in the attempt counts.

Several table cells average inside each source post first, then across posts. That makes the post the independent unit when several comments share a thread.

GPT-2 perplexity is how surprised a GPT-2 model is by the comment. Per text it is $\exp(\sum \text{NLL} / \sum \text{scored tokens})$, then averaged. Lower means more GPT-2-predictable, not more human. The 2026-08-15 paired freeze did not score GPT-2, unigram KL, JSD, BLEU, ROUGE, or BERTScore. Those columns appear only in Table 11.

The lexical-quality index is a 0–100 score from the comparison builder, version `lexical_quality_v2`. It is $100 \times (0.55 \text{ MATTR} + 0.30 \text{ bigram score} + 0.15 \text{ length sanity})$. MATTR is a moving-average type-token ratio with window 10. The bigram term penalizes a repeated two-word phrase. Length sanity is 1 for comments between 5 and 120 tokens and falls off outside that band. It is not a raw MATTR column.

Word count in both printed tables is the number of `[A-Za-z0-9’]+` tokens. The later publication runner uses a different word-boundary regex. Those lengths are not in these tables.

BERTScore F1 in Table 11 is the unrescaled figure from the 2026-07-30 report. It is overlap with a human reference comment, not a security score.

The LLM judge is not G-Eval. G-Eval is a separate 1–5 quality path with a matched human reference. Here we asked Codex Luna, at high reasoning, five blinded questions about already-generated comments. Lineups are seeded. The model is not deterministic. These scores are not a reader study, and they are not proof of authorship or security.

6.2 LLM-judge quality check

We took one our-method comment and one ZLG comment per post. The source freeze had 1,608 complete pairs from 339 posts. The judge set keeps, in sorted pair order, the first pair per post whose ZLG comment is unique in that pool. That yields 244 pairs from 244 posts, 244 unique texts on each arm. Ninety-five posts with no unique ZLG output were dropped, so repeated ZLG covers would not dominate the scores. The judge set is therefore a unique-ZLG subset, not a random sample of the 339 posts. The five checks reuse those pairs. They are not 2,928 independent generated samples. All 2,928 judge tasks completed with no task errors.

Figures 9–13 walk through one worked example each, all drawn from the same post so a reader can follow one thread across all five checks: 1lqu69t on r/news, “First immigration detainees arrive at Florida center in the Everglades.” The three candidate texts are our comment (“It feels like we are just spinning wheels by assuming opposition inevitably creates enemies without looking at the actual outcomes of these policies.”), ZLG’s comment (“He told them to leave and if not they’d be killed.”), and, where the criterion uses one, a held-out human comment (“They’re not calling it that. Also, what was Auschwitz, hmm?”). One post is an illustration, not a result; the bottom panel of each figure carries the actual 244-pair aggregate, and Table 10 collects all five.

6.2.1 Standout Detection. One generated comment is inserted into nine human comments from the same post. The judge names the inserted slot. Chance is 10%. Lower means the comment was harder to pick out as the extra one. Figure 9 shows this test on the worked example: the judge caught our comment at slot 1 with 93% confidence, calling it “unusually generic, abstract, and detached from the thread’s concrete facility details”; when ZLG’s comment was the one inserted instead, the judge named a different, unrelated comment and missed ZLG entirely. That is the harder direction for our method, not the flattering one. The judge prompt supplies the full post and all ten numbered comments and asks for JSON only:

Below is a discussion thread from r/{subreddit}, followed by 10 comments from that thread. Exactly one comment was machine-generated and inserted. Identify it. Judge conversational fit, specificity, stance, register, and habits; do not use length or polish as a cue.

Post title: {post_title}
 Post body: {post_body}
 Comments:
 {numbered_comments}

Return JSON only: {"inserted_index": <1-10>, "confidence": <0-100>,
 "runner_up_index": <1-10>, "evidence": "<short sentence>"}

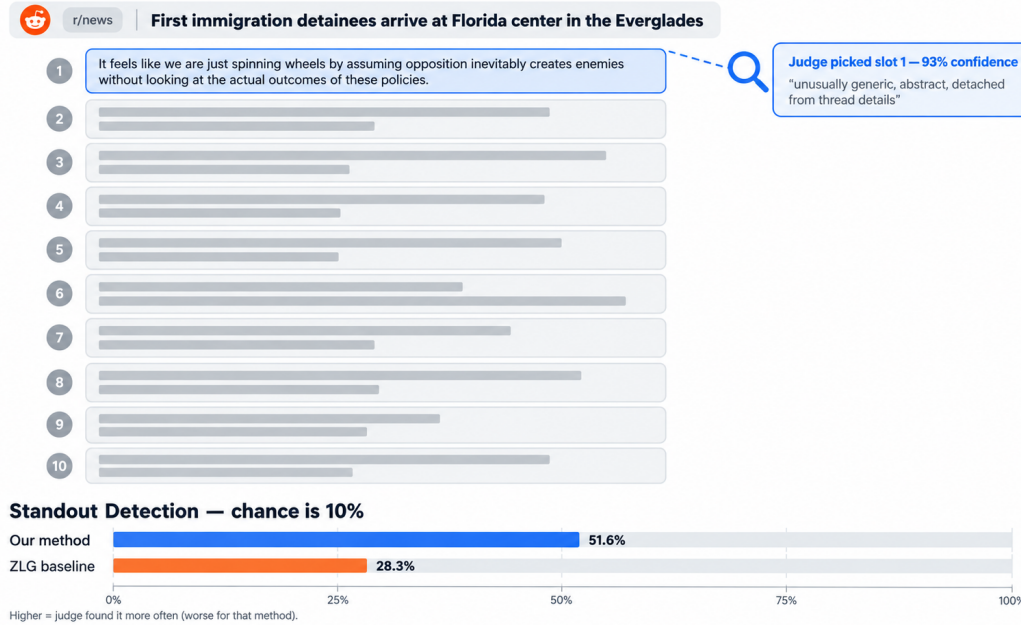


Fig. 9. Standout detection, worked example. The judge is shown the full thread and finds our inserted comment at 93% confidence; ZLG’s comment, inserted in a separate task, is missed. Across all 244 posts, ZLG is picked out less often (28.3% versus 51.6%), against a 10% chance baseline.

6.2.2 Weakest Text. One lineup holds a human comment, our comment, and the ZLG comment. The judge names the weakest of the three. Chance is 33%. Lower is better. The three pick rates share the same 244 lineups and sum to 100%. Figure 10 describes this test on the worked example: the judge ranked ZLG’s terse reply as most human-plausible, the human control’s comment second, and named our comment weakest, calling it “vague, polished abstraction [that] does not clearly engage the thread.” Individual rounds like this one do not average to the aggregate: over the full set, our comments are named weakest less often than ZLG’s (29.1% versus 47.5%), though still more often than the human control (23.4%), which is the honest floor no method should expect to beat by much. For each task, a seeded random generator keyed by the post and pair IDs shuffles the three replies. The permutation is reproducible for that task but varies across tasks. All six reply orders appeared across the 244 saved tasks, with 36 to 45 tasks per order. The prompt gives the thread and replies, then asks for a full ranking plus the weakest index:

Below is a discussion thread from r/{subreddit}, then three replies. Rank them from most human-plausible (1) to least (3), then name the weakest. Ignore length and ordering.

Post title: {post_title}

Post body: {post_body}

Existing comments:

{numbered_context_comments}

Replies: 1. {candidate_1}

2. {candidate_2}

3. {candidate_3}

Return JSON only: {"weakest_index": <1-3>, "ranking": [<index>, <index>, <index>], "evidence": "<short sentence>"}

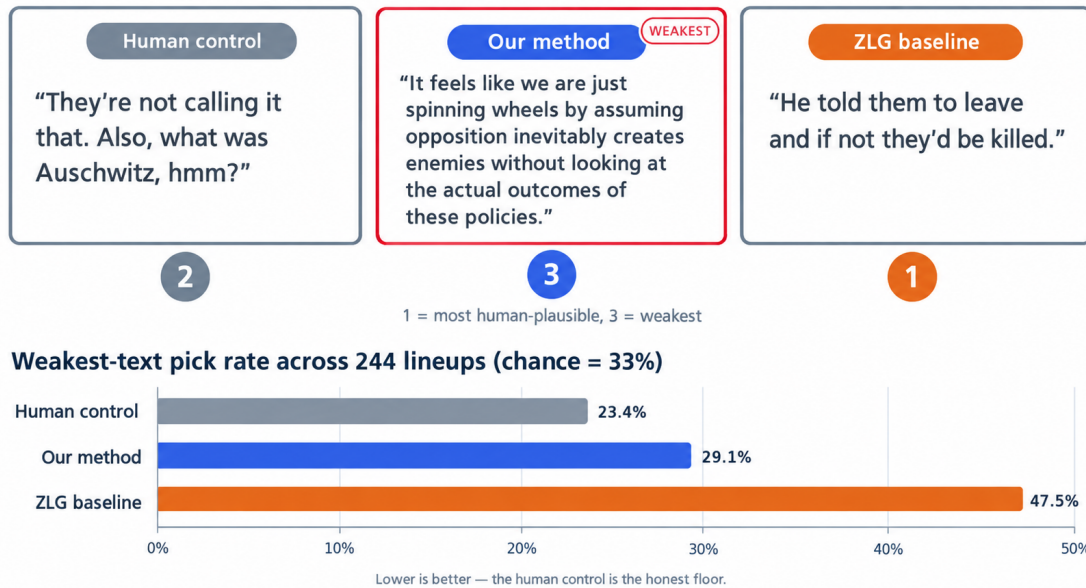


Fig. 10. Weakest-text pick, worked example. In this lineup the judge names our comment weakest; across 244 lineups the pattern reverses, with ZLG named weakest more often (47.5% versus 29.1%) than the human control (23.4%).

6.2.3 Suspicion AUROC. Each arm, including a held-out human comment, gets a 0–100 suspicion score. AUROC here is how often that method’s comments look more suspicious than the human comments. Ties count as 0.5. Chance is 0.5. Lower is closer to the human scores. The reported $p = 0.385$ tests the paired raw scores, not the AUROC gap. Figure 11 shows this test on the worked example, where the scores are unusually clean: our comment scored 82/100, flagged for four tells including `generic_could_fit_any_thread` and `register_mismatch`; ZLG scored 10/100, “no clear machine-generated signals”; the human control scored 3/100. The example’s wide score gap does not hold up across all 244 pairs: AUROC is 0.661 for our method versus 0.650 for ZLG, and a paired test of the raw suspicion scores does not show a clear difference ($p = 0.385$). Each text is scored alone, with no lineup and no knowledge of the other arms:

You moderate r/{subreddit}. Score the reviewed comment from 0 (genuine human) to 100 (machine-generated or edited). Do not treat bluntness, mistakes, or poor punctuation as evidence.

Post title: {post_title}

Post body: {post_body}

Recent comments:

{numbered_context_comments}

Comment: {candidate}

Select applicable tells from `generic_could_fit_any_thread`, `ignores_thread_specifics`, `register_mismatch`, `unnatural_phrasing`, `repetition`, `over_hedged`, `over_polished`, `contradicts_thread_facts`, `none`.

Return JSON only: {"suspicion": <0-100>, "tells": ["<tell>"], "evidence": "<short sentence>"}

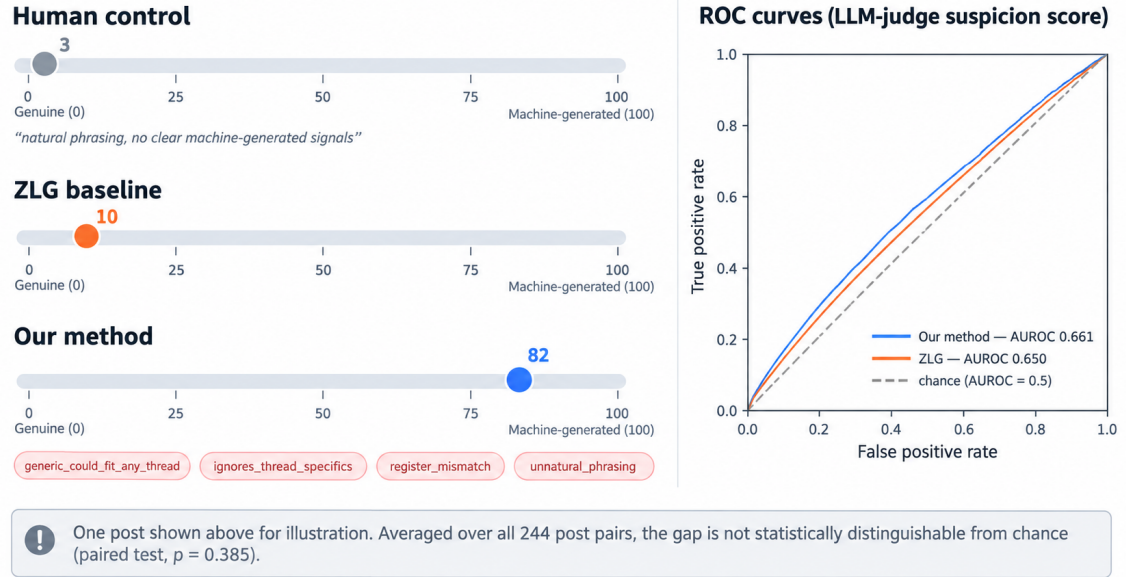


Fig. 11. Suspicion score, worked example. This post shows scores of 82 versus 10. Across 244 pairs, AUROC is 0.661 for our method and 0.650 for ZLG; the paired test of raw suspicion scores shows no clear difference ($p = 0.385$).

6.2.4 Thread Attribution. The judge picks which of four thread blurbs a comment belongs to. Chance is 25%. Higher means stronger thread-specific linkage. We report the rate. We did not run a significance test on this row. Figure 12 shows this test on the worked example: the judge correctly matched our comment back to its thread at 88% confidence, citing its engagement with “opposition to immigration enforcement policies.” It also matched ZLG’s comment to a thread, but the wrong one, at 68% confidence, because “He told them to leave and if not they’d be killed” reads generically enough to fit more than one immigration story. That pattern holds in aggregate: our method attributes more accurately than ZLG (89.3% versus 75.0%), which is not automatically a point in our favor here, since it means our comments carry more thread-specific content, not that they read more naturally. The comment is shown with no lineup context, only four candidate threads:

Below is a comment and four discussion threads. The comment belongs to exactly one thread.

Choose the most likely thread even if uncertain.

Comment: {candidate}

1. r/{sub_1} -- {title_1}

{snippet_1}

2. r/{sub_2} -- {title_2}

{snippet_2}

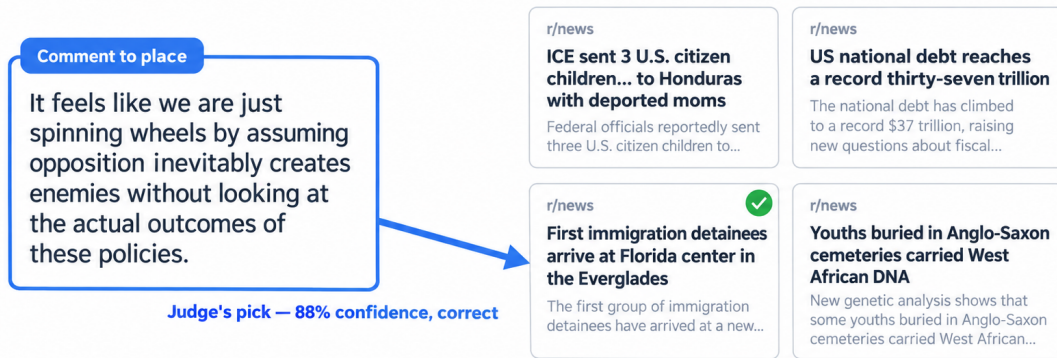
3. r/{sub_3} -- {title_3}

{snippet_3}

4. r/{sub_4} -- {title_4}

{snippet_4}

Return JSON only: {"thread_index": <1-4>, "confidence": <0-100>, "evidence": "<short sentence>"}



Thread attribution accuracy across 244 comments (chance = 25%)

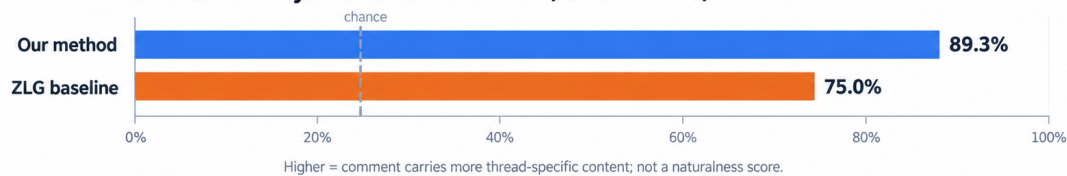


Fig. 12. Thread attribution, worked example. The judge correctly places our comment at 88% confidence; across 244 comments our method attributes more accurately than ZLG (89.3% versus 75.0%), a sign of thread-specific content, not naturalness.

6.2.5 Register-Fit. Using five genuine subreddit comments as examples, the judge scores style on a 1–5 overall field. The table cell is the arithmetic mean of that field. Higher is better. This is the only criterion with a paired per-post sign test in the scoring artifact. Figure 13 shows this test on the worked example: ZLG scored 4/5, “concise, direct, consistent with the thread’s plainspoken political commentary,” while our comment scored 2/5, “more formal and generic than the thread’s concrete, informal, profanity-laced discussion.” This example runs against the aggregate: over 244 posts our method scores higher on average (3.193 versus 2.791), with 106 posts favoring us, 67 favoring ZLG, and 71 ties (paired sign-test $p = 0.00374$), a real if modest effect, not a rout. Five real comments from the same subreddit anchor the judge’s sense of local style before it scores the candidate:

Five genuine $r/\{\text{subreddit}\}$ comments show the community register:

$\{\text{numbered_context_comments}\}$

Judge this comment’s style only: $\{\text{candidate}\}$

Score tone_formality , length_norm , mechanics , insider_knowledge , and overall 1 (out of place) to 5 (indistinguishable).

Return JSON only: $\{\text{"tone_formality": } <1-5>, \text{"length_norm": } <1-5>, \text{"mechanics": } <1-5>, \text{"insider_knowledge": } <1-5>, \text{"overall": } <1-5>, \text{"evidence": " <short sentence>"}\}$

Five genuine r/news comments shown as style anchors



Register-fit score across 244 posts (1-5 scale, higher is better)

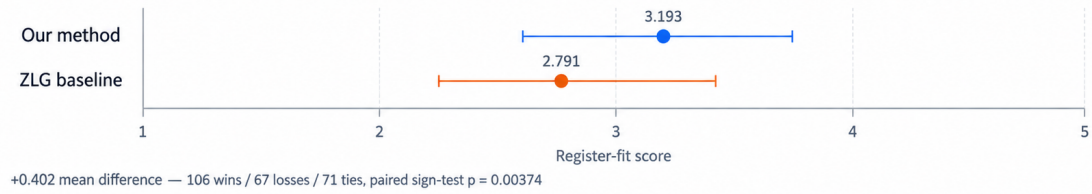


Fig. 13. Register-fit, worked example. ZLG scores higher on this post (4/5 versus 2/5), the reverse of the aggregate: over 244 posts our method's mean is higher (3.193 versus 2.791; paired sign-test $p = 0.00374$).

Table 10. Independent LLM-judge result on 244 unique-output post pairs (Codex Luna, high reasoning). Lower is better for standout, weakest-text, and suspicion AUROC. Higher is better for attribution and register-fit. Chance is 10%, 33%, 0.5, and 25% for the first four rows.

Criterion	Our method	ZLG	Note
Standout detection	126/244 (51.6%)	69/244 (28.3%)	ZLG picked less often in the 10-comment lineup
Weakest text picked	71/244 (29.1%)	116/244 (47.5%)	Human control 57/244 (23.4%)
Suspicion AUROC vs human	0.661	0.650	Paired raw scores close ($p = 0.385$)
Thread attribution	89.3%	75.0%	Rate only; no significance test
Register-fit (1–5)	3.193	2.791	+0.402 per post; 106/67/71; $p = 0.00374$

ZLG is harder to pick out of a same-thread lineup. Our comments are named weakest less often than ZLG's, though still more often than the human control. Register-fit favors our method on the paired sign test. Suspicion AUROC is close for both arms. None of this is a security result, and none of it uses the 244 unique-text count from the 2026-07-30 run.

6.3 How often generation succeeded

These counts are not the 244-pair judge set.

The run that fed the comparison requested 4,044 samples and recorded 2,543 successes and 1,501 failures (846 generation failures, 613 comment topic mismatches, 42 invalid JSON). Every recorded success reconstructed the compressed bitstring from sender-side audit metadata. That is an engineering check, not Bob reading a public thread alone. The

Table 11. Historical paired results over 100 source posts (2026-07-30). Lower is better for GPT-2 perplexity. Lexical quality is the run’s 0–100 index, not a raw MATTR column. BERTScore F1 is the unrescaled figure from that report.

Metric	Our method	ZLG	Paired result
Selection budget / ZLG reported payload	15.14	16.00	ZLG value higher; Holm-adjusted sign-test $p < 0.001$
GPT-2 perplexity	68.15	53.70	ZLG lower; $p = 0.0001$
Lexical quality index (0–100)	99.36	98.45	Ours higher; $p = 0.0015$
Word count	22.4	17.3	Ours longer; $p < 0.001$
BERTScore F1	0.842	0.848	Comparable in that report

campaign set `max_retries` to 1. The balanced profile default is 6 retries after the first attempt, which is what Section 4 describes.

The ZLG freeze that produced the 1,608-pair source file attempted 1,711 hides, accepted 1,610, and decode-verified 1,603 (96 quality-gate failures, 5 leakage-check failures, 0 harness skips). Our paired rows in that freeze are also 1,608, all reconstructed from sender audit metadata. A later completed ZLG file in the same run directory processed all 2,543 successes from that run and accepted 2,340. The judge did not use that larger file.

On the 1,608-pair freeze, means over 339 posts give an available fixed-width selection budget of 13.42 per comment for our method and reported ZLG payload of 13.90, with the ZLG value higher on 230 posts (Holm-adjusted sign-test $p = 1.49 \times 10^{-14}$). The comparison does not measure independently recovered project-side payload. Mean word counts are 23.86 versus 15.91. The lexical-quality index is 99.37 versus 96.93. GPT-2 perplexity and unigram KL/JSD were not scored on this paired freeze.

The same freeze also stores a self-consistency score (mean MiniLM cosine to other same-post outputs of that method), a shared naturalness gate (ours 1603/1608 passed, ZLG 1509/1608), and a repetition ratio. Self-consistency has no preferred direction for this comparison, so it is not a ranking row.

6.4 Historical configuration comparison

Table 11 is the 2026-07-30 `context_weighted_v2` run, not the 2026-08-15 freeze. Our method attempted 627 samples and succeeded on 554 (88.4%). Failures were 70 comment topic mismatches and 3 generation failures. ZLG was fed those 554 successes in a capacity-matched setting with EGS parameters (threshold = 0.01, temperature = 0.7, alpha = 1.0, `max_bpw` = 2). The original writeup counts 304 accepted of 554 (54.9%).¹ The metric table uses those accepted pairs, then averages over 100 source posts.

The candidate topic list was capped at 32, so the comment topic component contributed at most 5 bits to the available fixed-width selection budget per frame. Project-side rows reconstructed the compressed bitstring from sender-side audit metadata. The research pool was 167 posts, 35 of them sampled ten times, and 244 of the 304 our-method texts were unique. That 244 is not the judge subset.

On that configuration, the reported ZLG value exceeds our available fixed-width selection budget, and ZLG has lower GPT-2 perplexity. The bit comparison is not a comparison of independently recovered useful payload. Our replies are longer and score slightly higher on the lexical-quality index. The numerical gap is narrow because the candidate topic list was capped at 32. It is not a general upper bound on either system.

¹A 2026-07-31 recode of the same run treats 94 cover-sentence extraction skips as harness faults, leaving 460 server invocations, 304 accepted (66.1%), and 156 genuine ZLG failures (133 quality-gate, 22 leakage, 1 HTTP 500). The metric table still uses the 304 accepted pairs.

6.5 What these comparisons do not show

The judge result is about the unique-ZLG-output subset, one pair per post. The historical table is one comment topic capacity configuration and one baseline service configuration, with reused posts and audit-assisted reconstruction on our side. KL/JSD, synthetic detection, and grouped passive-detector ROC-AUC are computed by later runners and are not in the 2026-08-15 paired freeze. A dedicated steganalysis and editing-attack evaluation remains future work. This paper does not claim a security advantage for either method.

7 CONCLUSION

This work hides a payload in a threaded discussion by choosing a reply target and a comment topic, then asking an LLM to write an ordinary comment. Capacity is whatever those two lists uniquely decode. The design does not insert bits into next-token sampling, and it does not need white-box logits.

An independent LLM-judge run on the 244-post unique-ZLG subset found ZLG harder to pick in a same-thread lineup (28.3% versus 51.6% standout detection) and our method higher on register-fit (3.193 versus 2.791, paired sign-test $p = 0.00374$). The source campaign, run with one retry rather than the default six, succeeded on 2,543 of 4,044 attempts, all by audit-assisted reconstruction from sender-side metadata. A historical 2026-07-30 configuration, averaged over 100 posts, reported an available fixed-width selection budget of 15.14 bits/comment for our method versus 16.00 reported bits for ZLG, and GPT-2 perplexity of 68.15 versus 53.70 (ours versus ZLG), with 554 of 627 project-side attempts succeeding. These bit figures do not measure independently recovered project-side payload; the ZLG reveal used sender token IDs and externally supplied payload length. Those two artifacts answer different questions. Neither is a blind, capacity-matched ranking, and neither establishes resistance to an adaptive steganalyst. Future work should freeze a symmetric manifest, score GPT-2 and divergence on that freeze, decode without sender audit files, and test dedicated steganalysis and editing attacks.

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